Chemistry: CYL1010

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Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

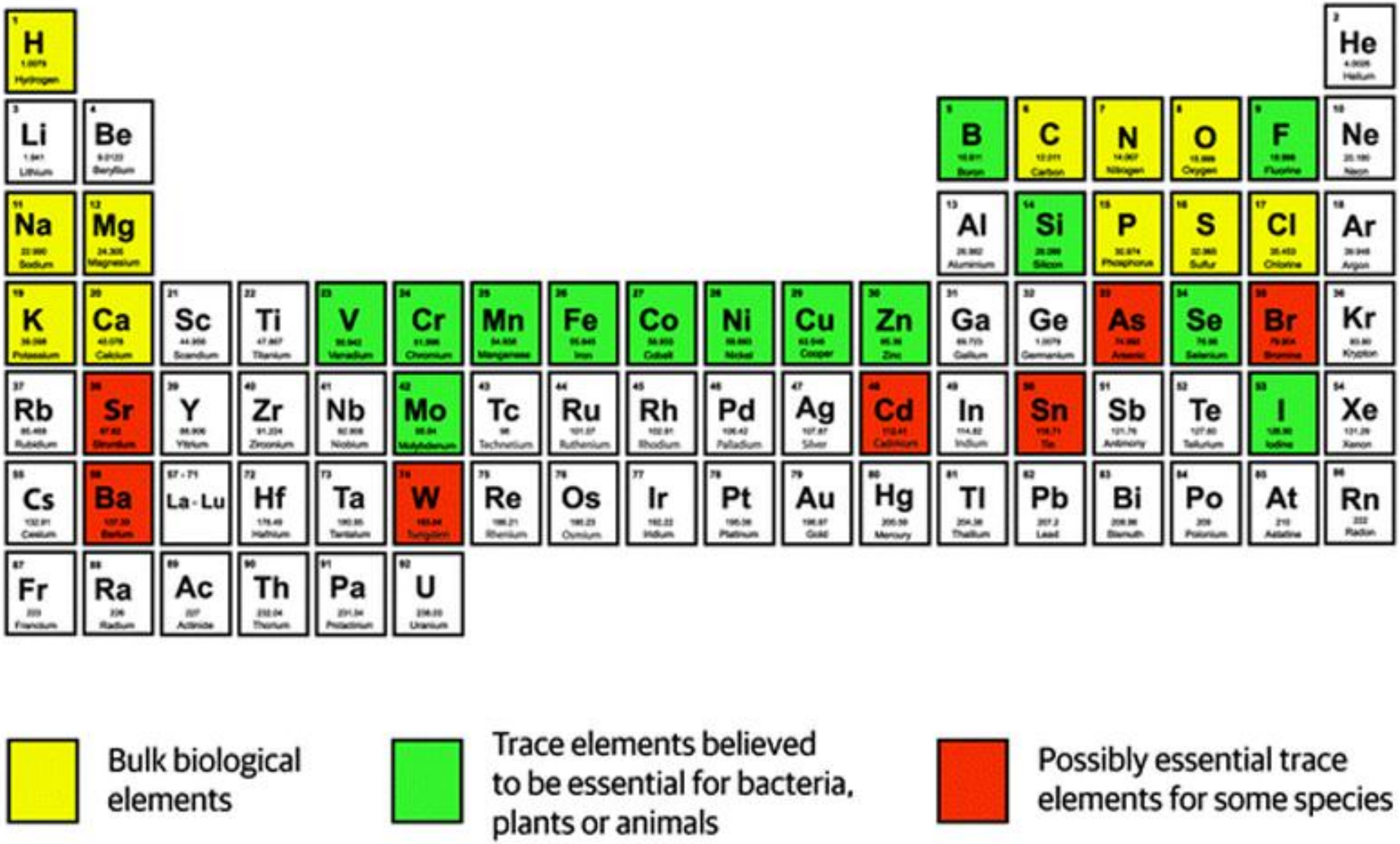
Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

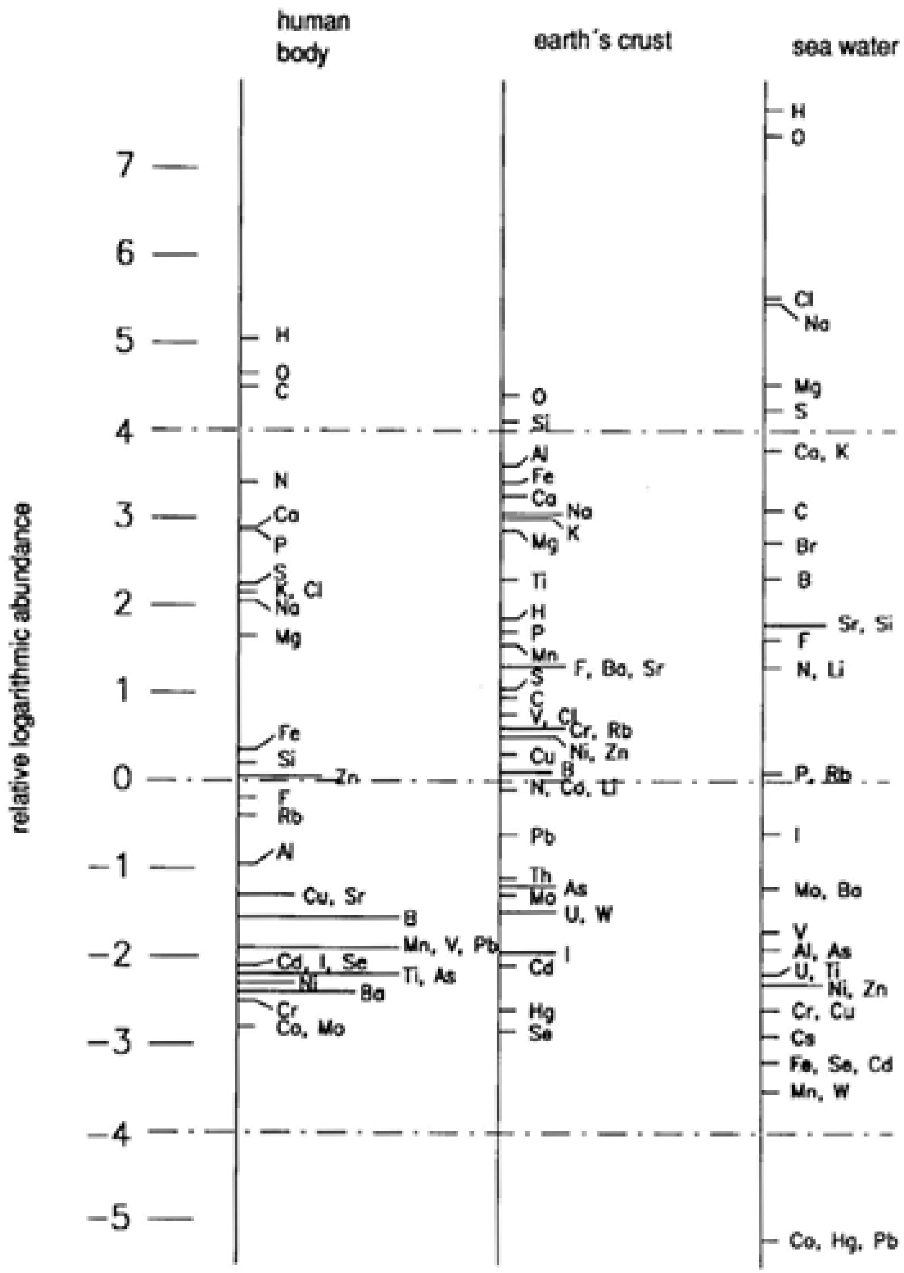
Bioinorganic chemistry

Bioinorganic chemistry is the study of the 'inorganic' elements as they are utilized in biology. The main focus is on metal ions, where we are interested in their interaction with biological ligands and the important chemical properties, they are able to exhibit and impart to an organism. These properties include ligand binding, catalysis, signalling, regulation, sensing, defence, and structural support.

- Study of the structural and functional aspects of metal bound species, such as proteins and nucleic acids in biological systems
- Metal ion transport and storage
- Metallohydrolase enzymes (peptidases)
- Metal-containing electron transfer proteins
- Oxygen transport and activation proteins
- Oxidation and hydroxylation (oxidases)
- Hydrogenases and transferases
- Enzymes involved in nitrogen metabolism pathways

Periodic Table Relevant to Life





Element	Sea Water (M) x 10^{-8}	Human Plasma (M) x 10^{-8}
Fe	0.005-2	2230
Zn	8.0	1720
Cu	1.0	1650
Mo	10	1000
V	4.0	17.7
Mn	0.7	10.9
Cr	0.4	5.5
Ni	0.5	4.4
Co	0.7	0.0025

Bertini, I.; Gray, H. B.; Lippard, S. J.; Valentine, J. S. Bioinorganic Chemistry; University Science Books: Sausalito, CA, 1994.

How nature has chosen these elements?
Criteria for the selection of elements

Elemental abundance is not ONLY the determining factor

- Solubility of the element
- Charge type/Oxidation state
- Ionic Radius
- Ligating atoms
- Preferential coordination geometry
- Spin-pairing stabilization
- Kinetic reactivity and other controls
- Thermodynamic aspects
- Chemical reactivity

- **Structure**
  - Bone and teeth
  - Cell membranes
  - DNA and RNA structure
  - Protein, including enzyme conformation

- **Electron transfer (Redox reactions)**
  - Fe, Cu, Mn, Mo, Ni, Co

- **Metabolism (Degradation of organic molecules)**

- **Activation of small molecules (O_2, CO_2)**

Bulk or Constituent Elements:

H, O, C, N, Ca, P, Na, K, S, Cl

Trace Elements:

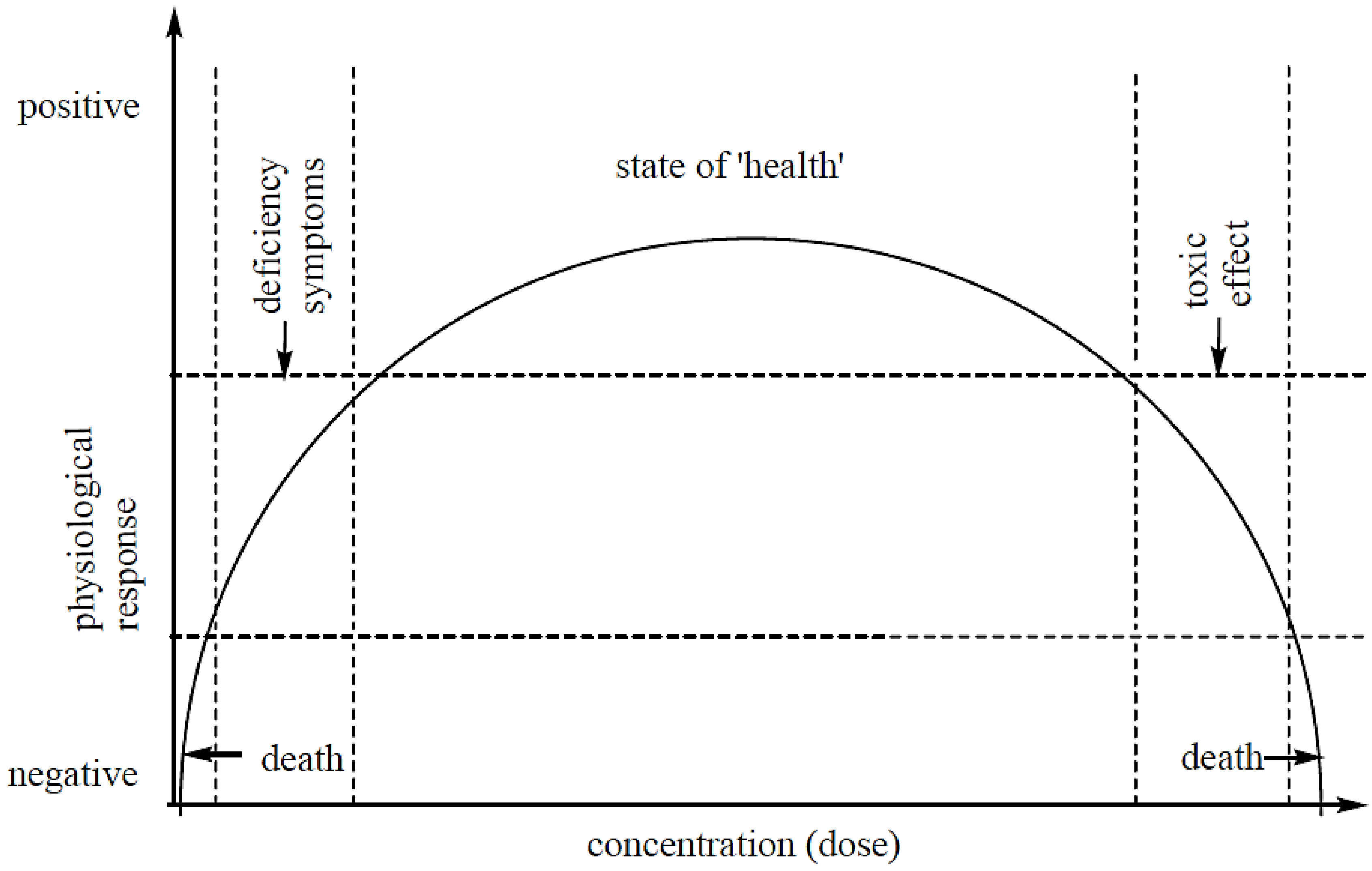
Mg, Si, F, Fe, Zn, B, Rb, Sr, Br, Cu

Ultra Micro Trace Elements:

V, Li, Se, Mn, Ba, Ge, As, Ni, Mo, Cd, I, Sn, Cr, Pb, Co

Metal	*Function*	*Typical deficiency symptoms*
Sodium	Charge carrier; osmotic balance	
Potassium	Charge carrier; osmotic balance	
Magnesium	Structure; hydrolase; isomerase	muscle cramps
Calcium	Structure; trigger; charge carrier	retarded skeletal growth
Vanadium	Nitrogen fixation; oxidase	
Chromium	Unknown, possible involvement in glucose tolerance	diabetes symptoms
Molybdenum	Nitrogen fixation; oxidase; oxo transfer	retardation of cellular growth; propensity for caries
Tungsten	Dehydrogenase	
Manganese	Photosynthesis; oxidase; structure	infertility; impaired skeletal growth
Iron	Oxidase; dioxygen transport and storage; electron transfer; nitrogen fixation	anemia; disorders of the immune system
Cobalt	Oxidase; alkyl group transfer	pernicious anemia
Nickel	Hydrogenase; hydrolase	growth depression; dermatitis
Copper	Oxidase; dioxygen transport; electron transfer	artery weakness; liver disorders; secondary anemia
Zinc	Structure; hydrolase	skin damage; stunted growth; retarded sexual maturation

Essential Element Dosage and Physiological Response



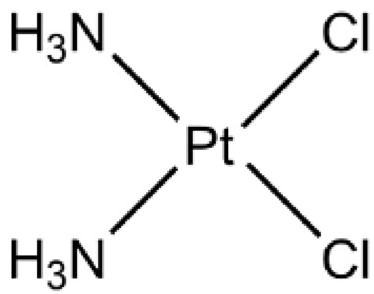
Why does biology utilize transition metals?

- Transition Metals are extremely good catalytic active sites in enzymes, because they:
  - are stable in a variety of geometries and CN
  - have multiple coordination sites
  - Are stable in a variety of oxidation states
  - Are able to change the reactivity of ligands
  - Have "weak" coordinate bonds (where needed)
  - Are capable of stabilizing intermediates

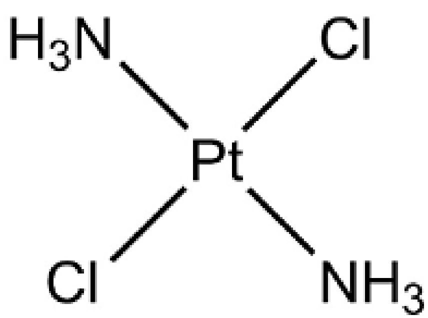
Why does it matter what ligands are attached to the metal ?

- Tune Redox Properties
- Assist in stabilization of multiple oxidation states of transition metals
- Lability/Stability

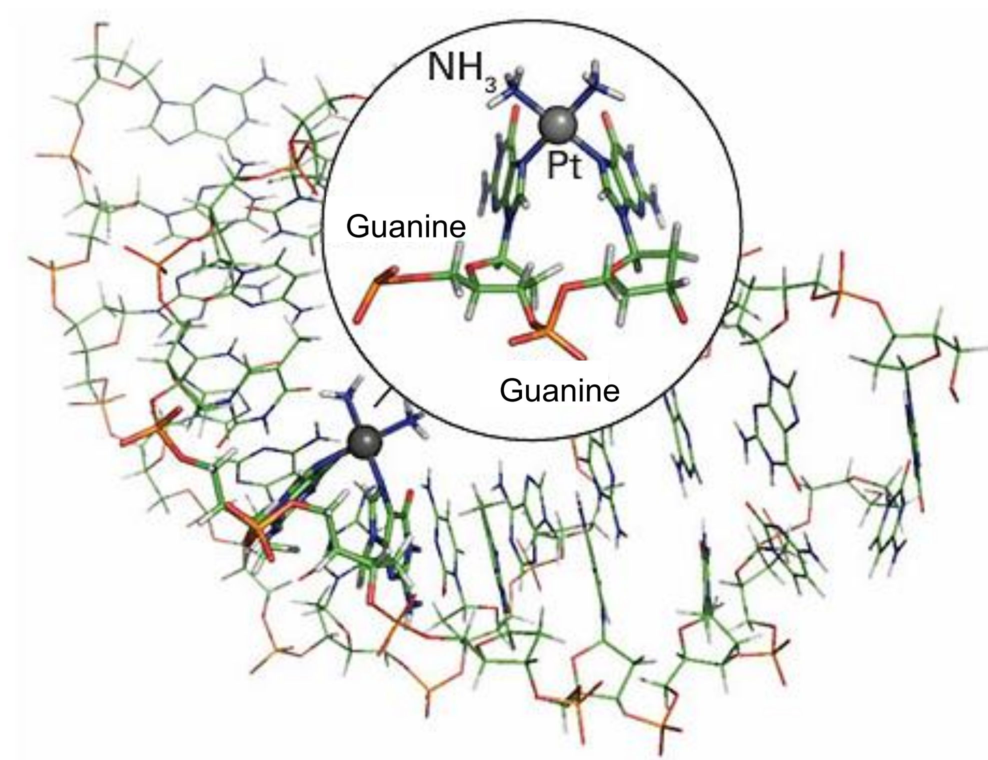
cis- and *trans*- Diamminedichloroplatinum (II)



Cisplatin



Transplatin



Cis-Platin is a highly effective chemotherapy drug for cancers due to its ability of cross-linking with DNA and inhibiting to copy the DNA by enzymes

Terms and Definitions in Bioinorganic Chemistry

Active center: Location in an enzyme where the specific reaction takes place.

Apo-enzyme: An enzyme that lacks its metal center.

Charge-transfer complex: An aggregate of two or more molecules in which charge is transferred from a donor to an acceptor.

Cluster: Metal centers grouped close together which can have direct metal bonding or through a bridging ligand, e.g. ferredoxin

Cofactor: An organic molecule or ion (usually a metal ion) that is required by an enzyme for its activity. It may be attached either loosely (coenzyme) or tightly (prosthetic group).

Cooperativity: The phenomenon that binding of an effector molecule to a biological system either enhances or diminishes the binding of a successive molecules, e.g. hemoglobin.

Terms and Definitions in Bioinorganic Chemistry

Entatic state: A state of an atom or group which has its geometric or electronic condition adapted for function. Derived from entasis (Greek) meaning tension.

Enzyme: A macromolecule that functions as a biocatalyst by increasing the reaction rate.

Heme: A near-planar coordination complex obtained from iron and dianionic porphyrin.

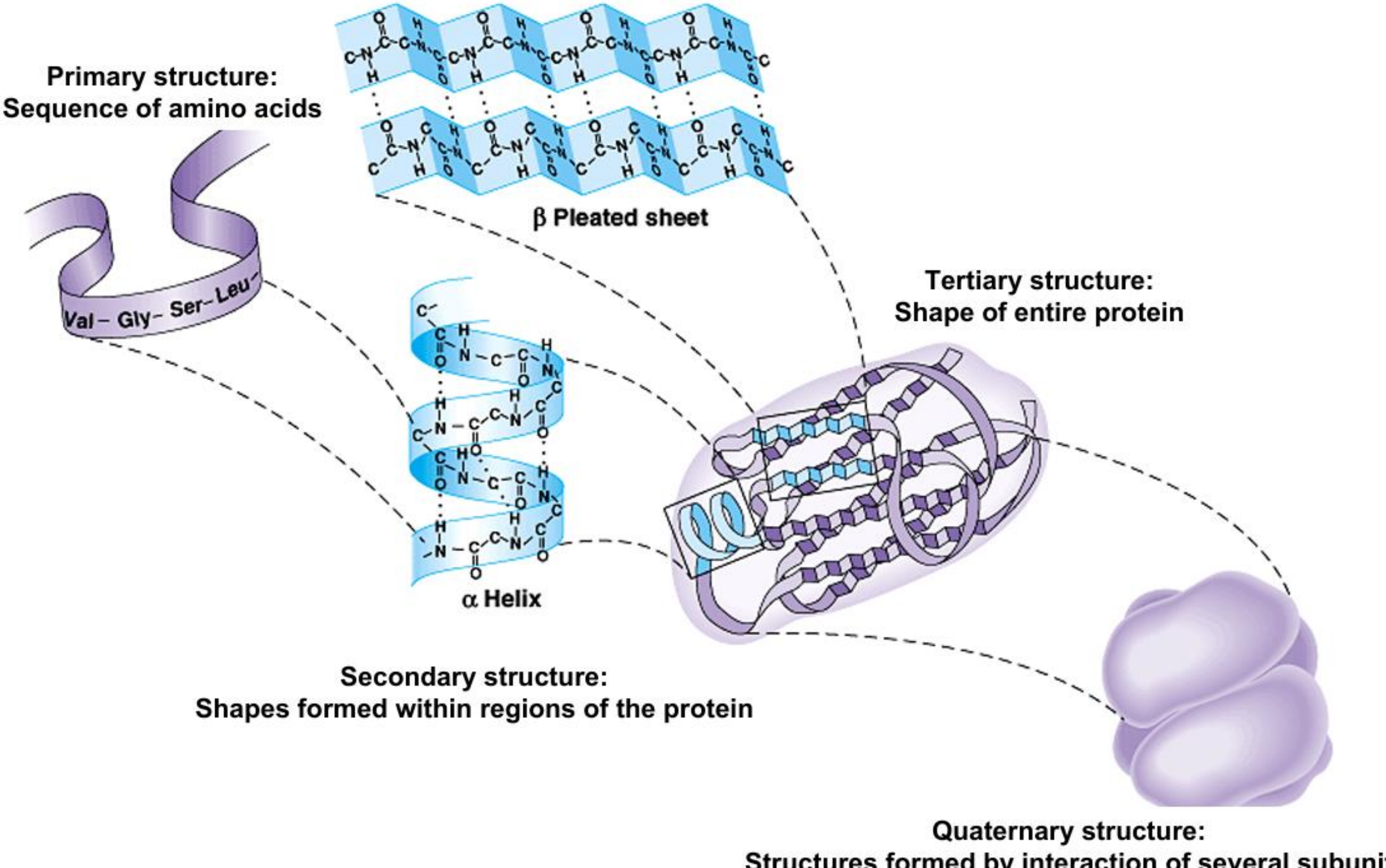
Hemoglobin: A dioxygen-carrying heme protein of red blood cells.

Holoenzyme: An enzyme containing its characteristic prosthetic group(s) and/or metal(s).

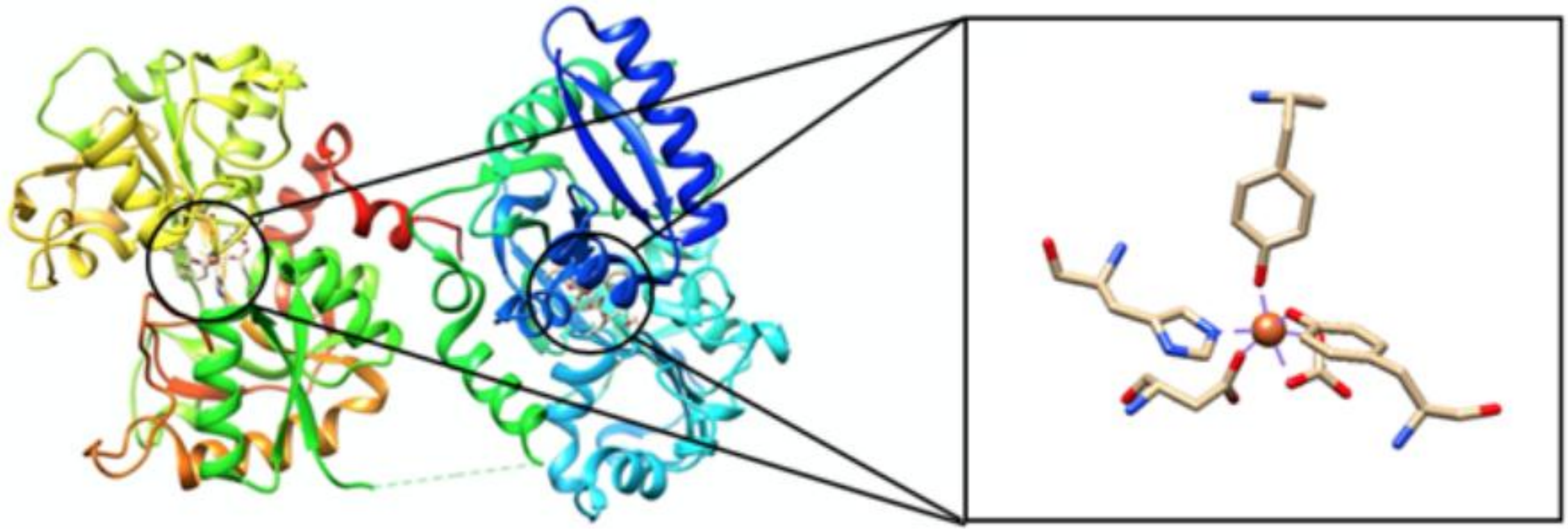
Ion channel: Enable ions to flow rapidly through membranes in a thermodynamically downhill direction after an electrical or chemical impulse.

Ionophore: A compound which can carry specific ions through membranes.

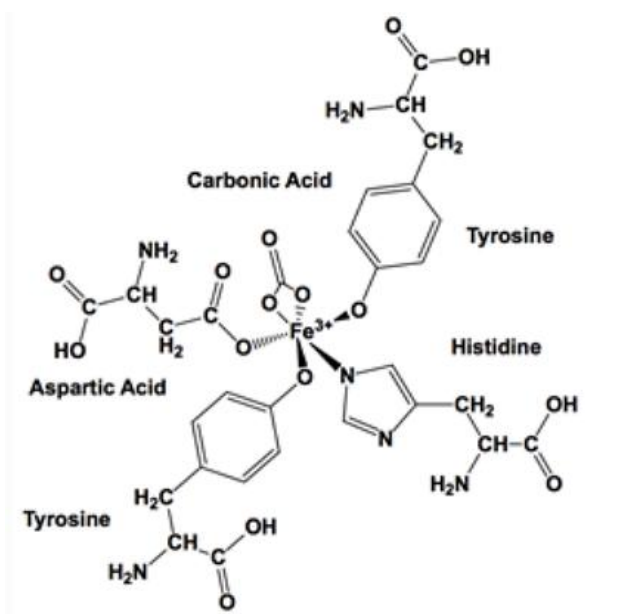
Ion pumps: Enable ions to flow through membranes in a thermodynamically uphill direction by the use of an energy source. They open and close upon the binding and subsequent hydrolysis of ATP, usually transporting more than one ion towards the outside or the inside of the membrane.


Primary structure:
Sequence of amino acids
Val – Gly – Ser – Leu
β Pleated sheet
α Helix
Secondary structure:
Shapes formed within regions of the protein
Tertiary structure:
Shape of entire protein
Quaternary structure:
Structures formed by interaction of several subunit

Human Serum Transferrin

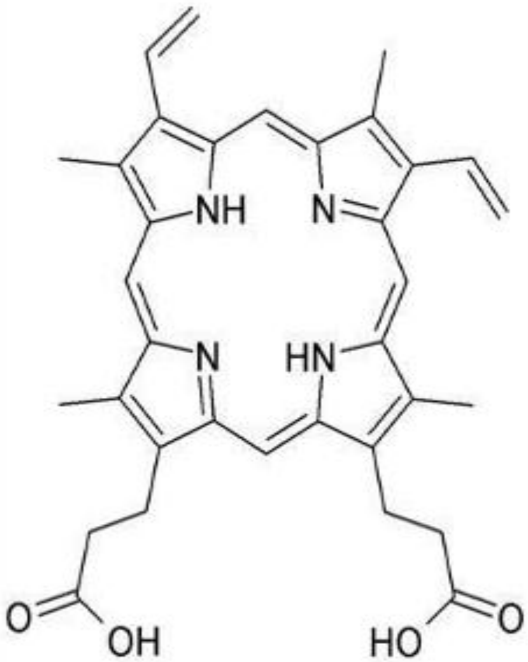


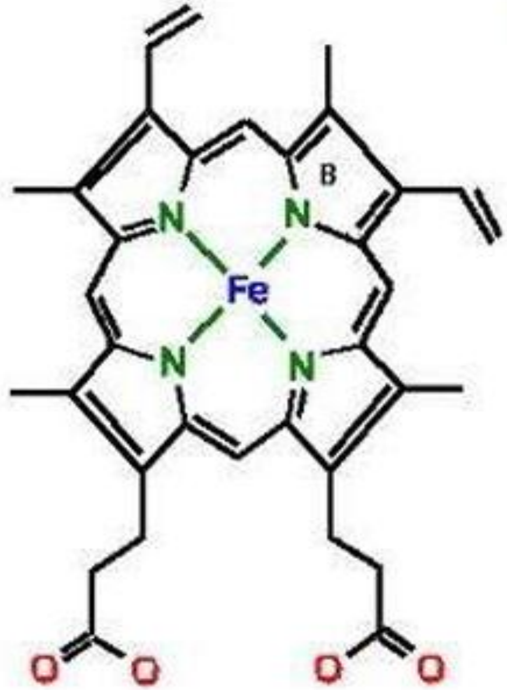
- This protein transports iron ions to various organs of the body
- Picks up iron from the storage protein, viz., ferritin, goes through the blood and delivers at the organ tissue
- During pick up and delivery iron is in +2; and during transport and storage it is in +3



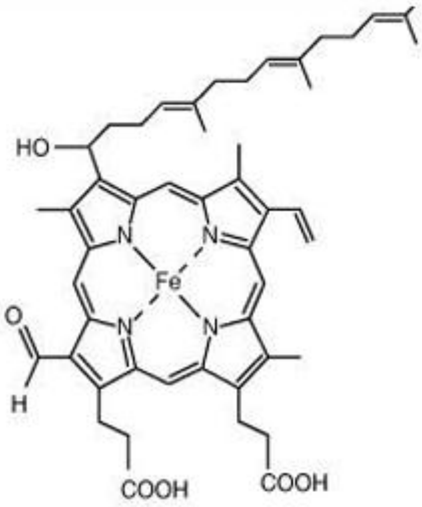
Porphyrin


3
5
7
2
4
6
8
NH
N
1
21
22
9
20
10
24
23
19
N
HN
11
18
12
16
14
17
15
13

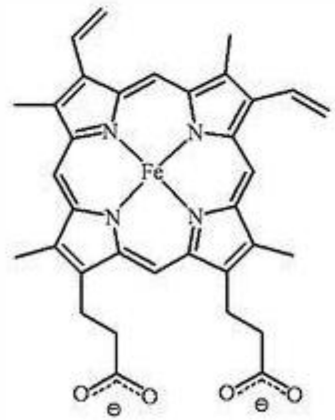

NH
N
N
HN
O
OH
HO
O


B
N
N
Fe
N
N
O
O
O
O

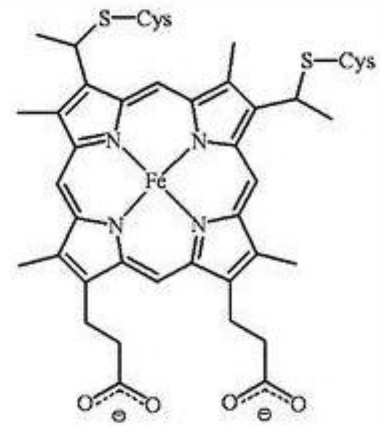
Heme and Haemoglobin



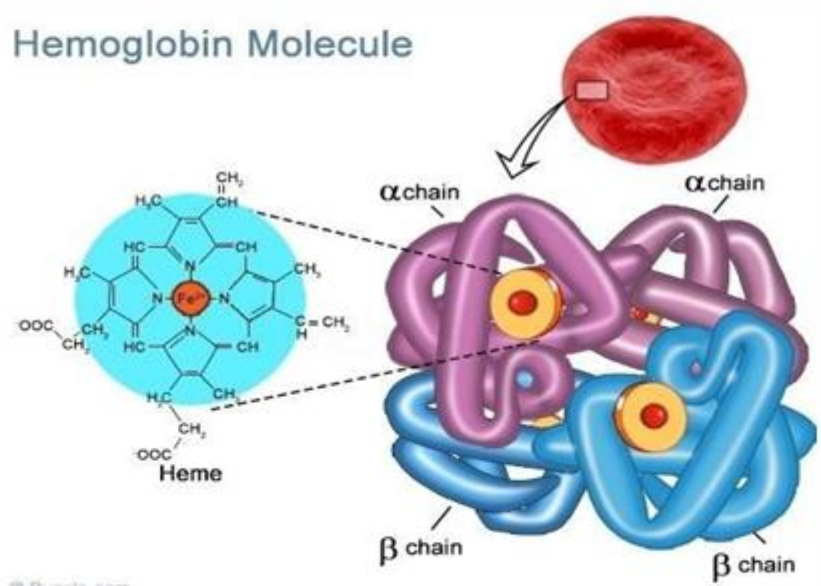
Heme A



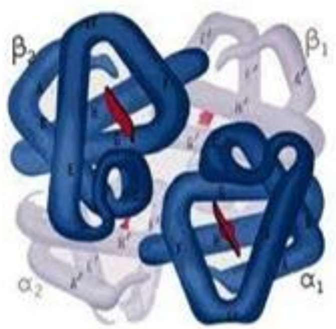
heme *b*



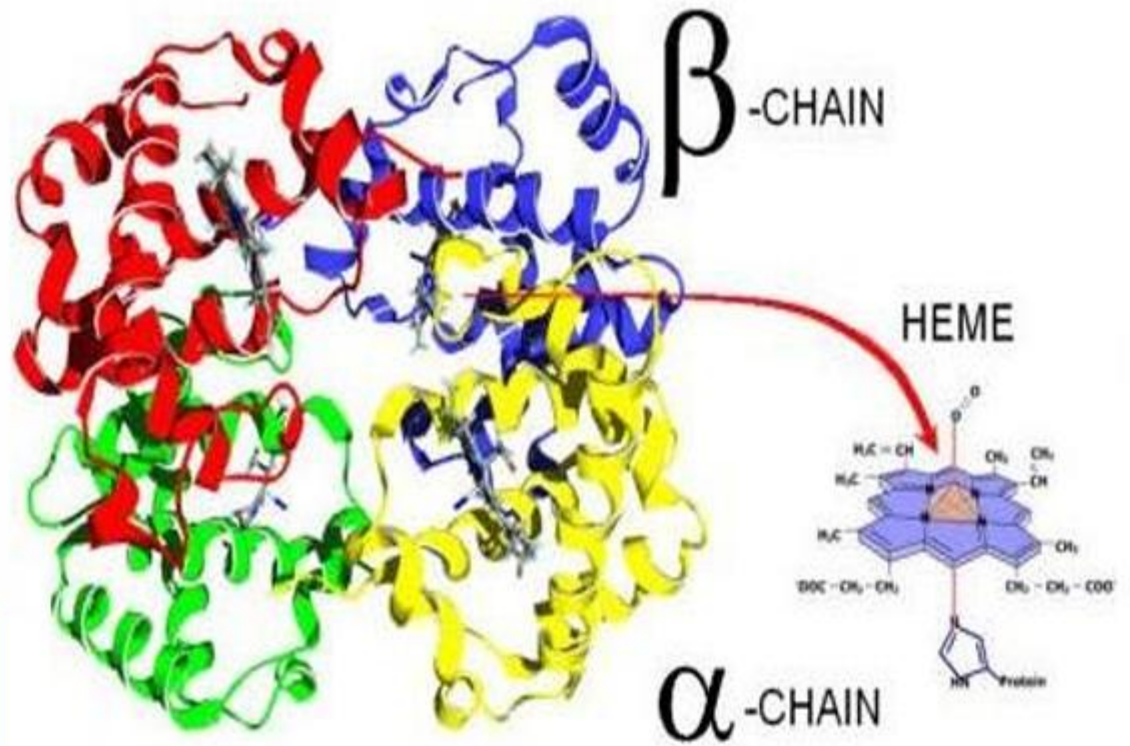
heme *c*



Hemoglobin Hb



Four units of Hb



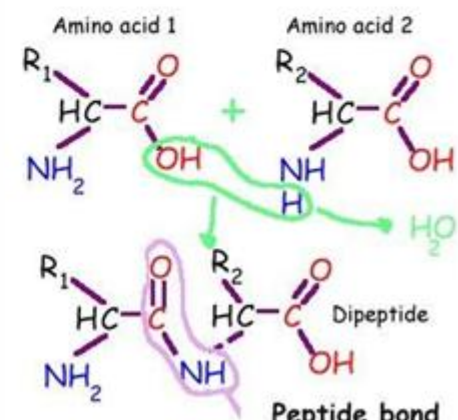
3 major types of Hb

Hb A (Adult)

Hb F (Fetal)

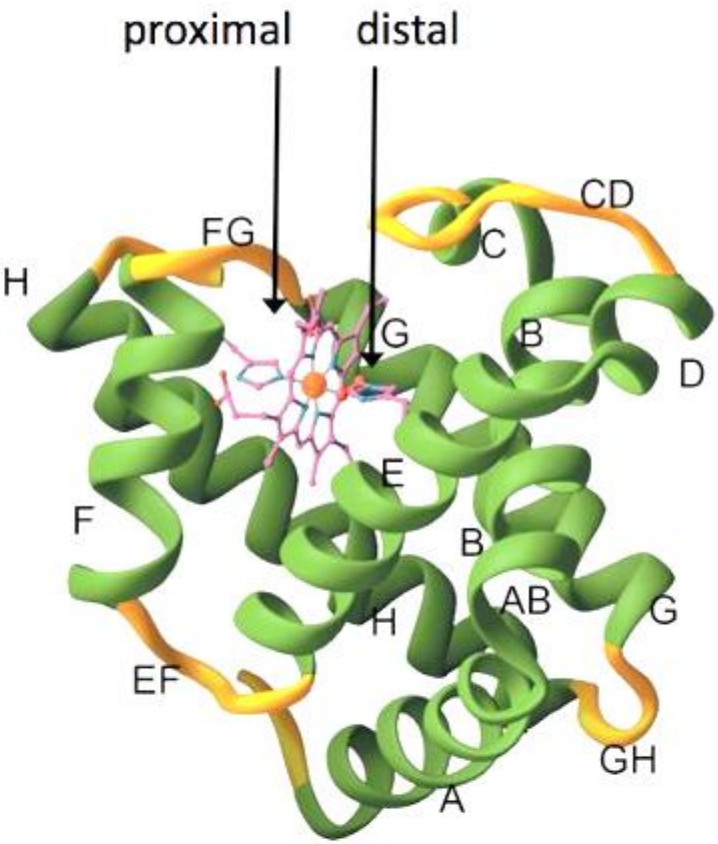
Hb S (Sickle cell)

Hb is tetramaric protein
M.W = 64500 Dalton

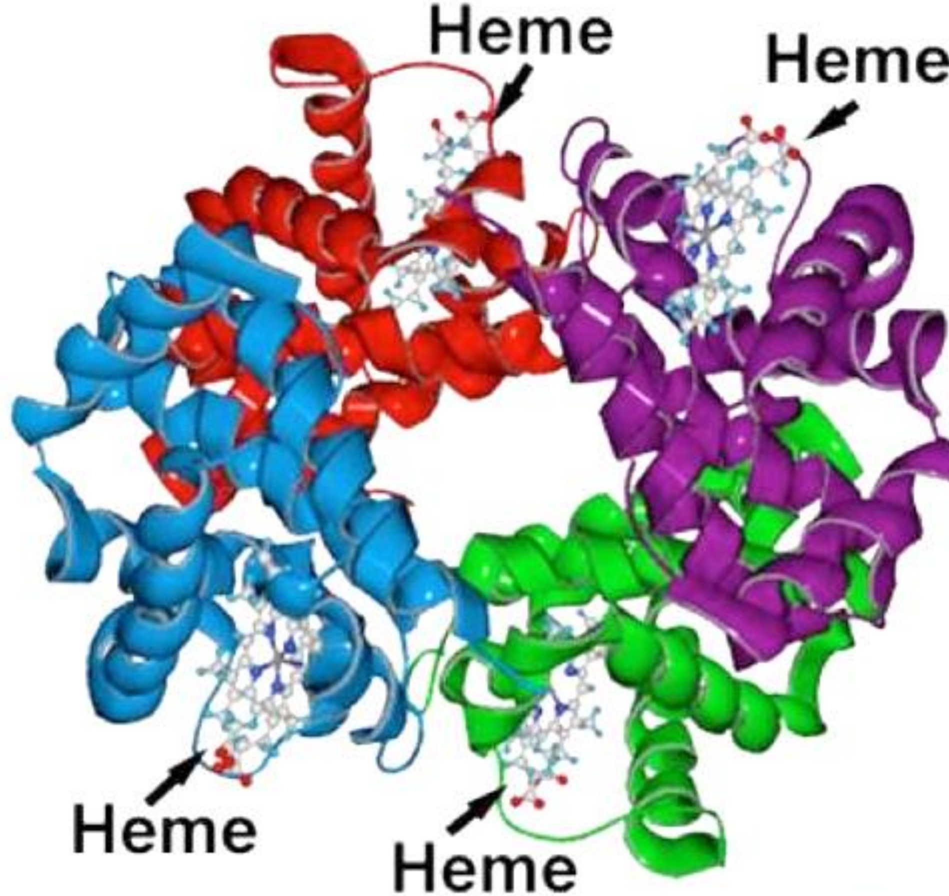


α Chain = 141 amino acid
β Chain = 146 amino acid

2α and 2β peptide chain are interlinked through H bonded interaction (COO^-…..NH)



V&V p 244 Fig. 8-39b Ribbon Structure of Myoglobin



www.animalscience.ucdavis.edu

Oxygenated Myoglobin

monomeric

Hemoglobin

Tetrameric

Myoglobin – O_2 storage; Hemoglobin – O_2 transport

Role of the protein in case of hemoglobin

Binding pocket of O_2 in protein:

Prevent 2-e reduction

Prevent μ-oxo dimer formation

Stabilizing PFe(II)…O_2 complex

Bent O_2 geometry

Binding of CO vs. O_2

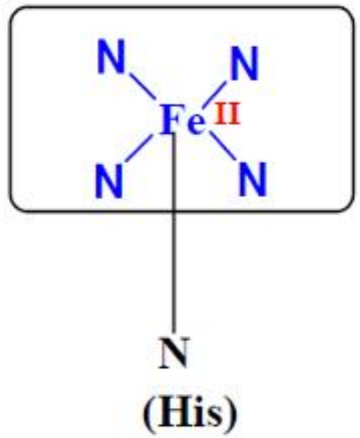
Thermodynamics vs. Kinetics -- Role of the protein

The problem:

$$Fe^{II}P + O_2 \rightleftarrows Fe^{III}P{-}O_2^{-} \xrightarrow{Fe^{II}P} PFe^{III}{-}\ddot{O}{-}\underset{\cdot\cdot}{O}{-}Fe^{III}P$$

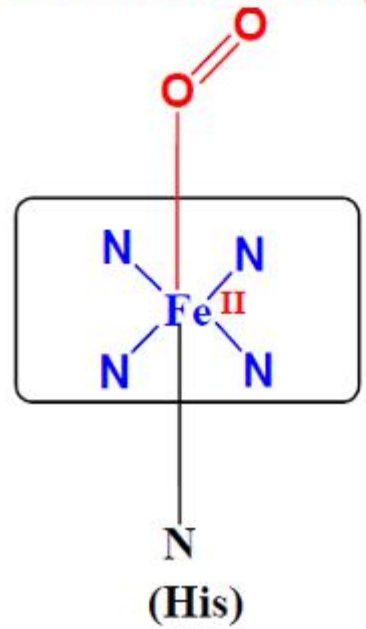
$$\underset{\text{ferryl}}{2PFe^{IV}{=}\ddot{O}:} \xrightarrow{Fe^{II}P} \underset{\text{μ-oxo, “dimer”}}{PFe^{III}{-}O{-}Fe^{III}P}$$

Oxy- and de-oxy forms of Hemoglobin



High Spin
paramagnetic
$t_{2}g^{4}\ e_{g}^{2}$

Deoxyhemoglobin



Low Spin
diamagnetic
$t_{2}g^{6}\ e_{g}^{0}$

Oxyhemoglobin

Deoxyhemoglobin is the form of hemoglobin without the bound oxygen. The oxyhemoglobin has significantly lower absorption (660 nm) than deoxyhemoglobin (940 nm). This difference is used for measurement of the amount of oxygen in patient's blood by pulse oximeter.

What happens when O_2 binds to Hemoglobin

The size of Fe^{2+} increase by 28% on going from

Low spin (oxyhemoglobin) (0.61 Å)

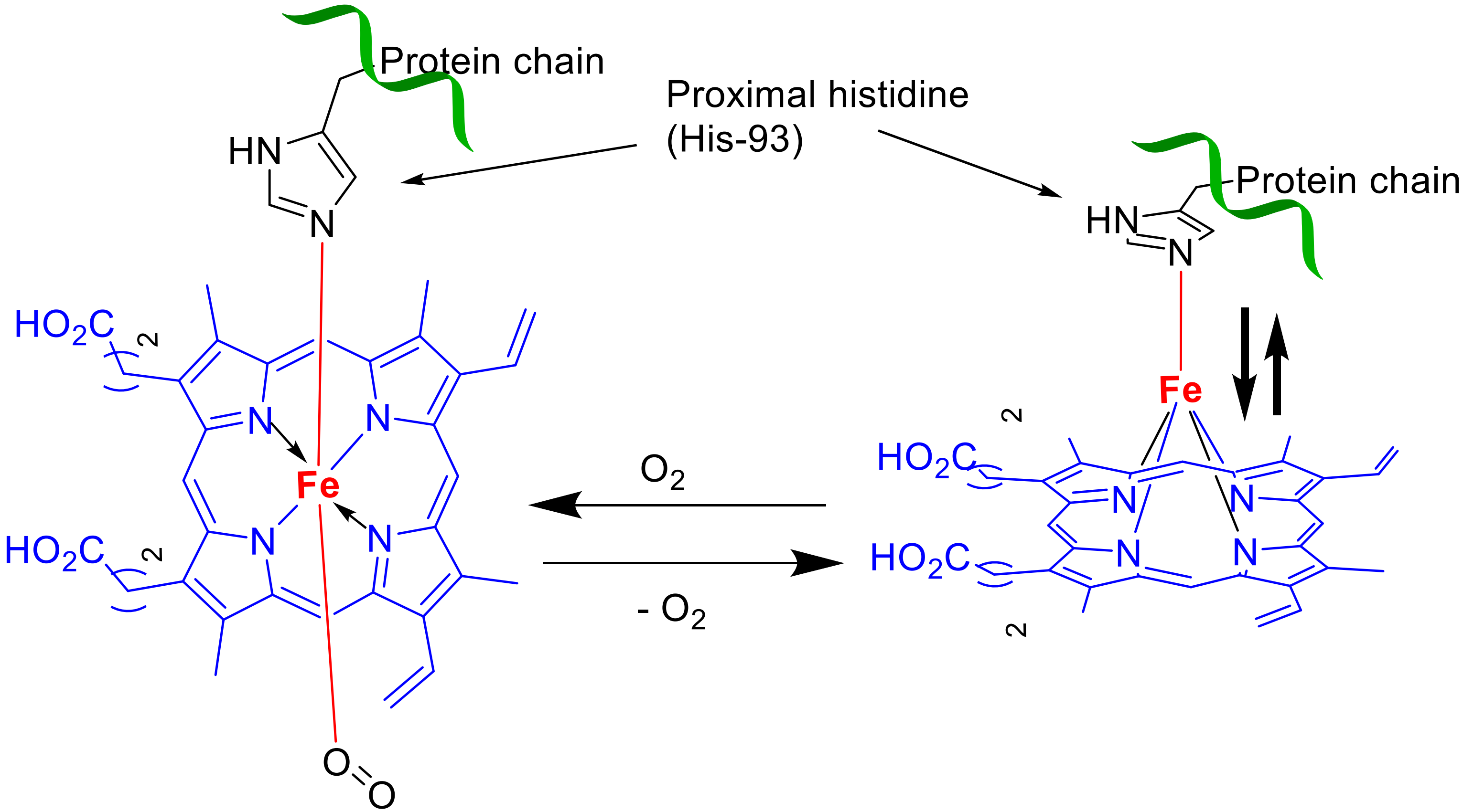
$\downarrow$ to

High spin (Deoxyhemoglobin) (0.78 Å)

The Fe^{2+} in deoxyhemoglobin is too large to fit in the ring and is situated (0.7-0.8)A° above the ring

Thus, presence of O_2 changes the electronic arrangement of Fe^{2+} and distorts the shape of the complex

The globular protein prevents the irreversible oxidation of Fe(II) to Fe(III)



Oxymyoglobin/Oxyhemoglobin
Six coordinated iron atom,
Low spin, in the palne of heme
R-state

Deoxymyoglobin/deoxyhemoglobin
Five coordinated iron atom,
High spin, out of the palne of heme,
T-state

Co-ordination of one O_2 leads to conformational changes in the protein chain leading to facilitate co-ordination of O_2 by other 3-sub-unit

Cooperativity

When O_2 binds to one sub-unit → Fe^{2+} contracts, moves into plane of porphyrin ring

↓

moves the histidine attached to it → triggers conformational changes in the globin chain

↓

translated through H-bond network → Enhances the ability of other three units to bind O_2

↓

This phenomenon is called cooperative effect

Transport of dioxygen

$$Hb + O_2 \underset{}{\overset{k_1}{\rightleftharpoons}} HbO_2$$

$$HbO_2 + O_2 \overset{k_2}{\rightleftharpoons} Hb(O_2)_2$$

$$Hb(O_2)_2 + O_2 \overset{k_3}{\rightleftharpoons} Hb(O_2)_3$$

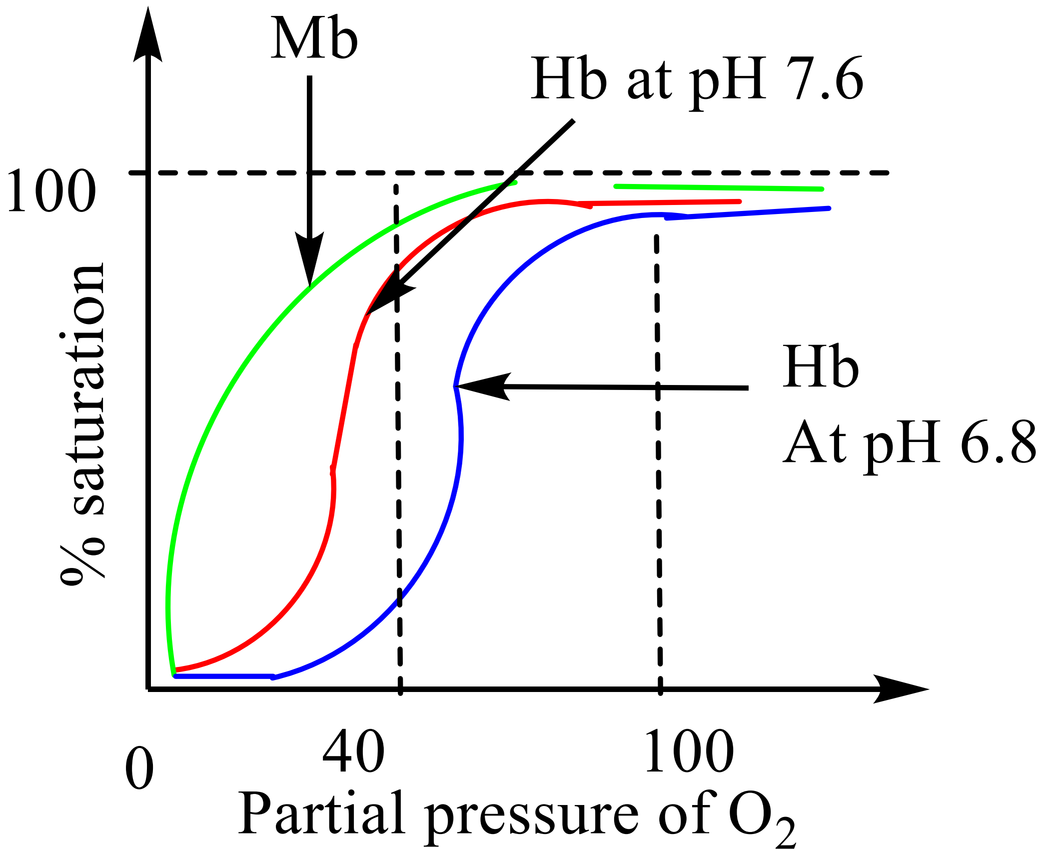
$$Hb(O_2)_3 + O_2 \overset{k_4}{\rightleftharpoons} Hb(O_2)_4$$

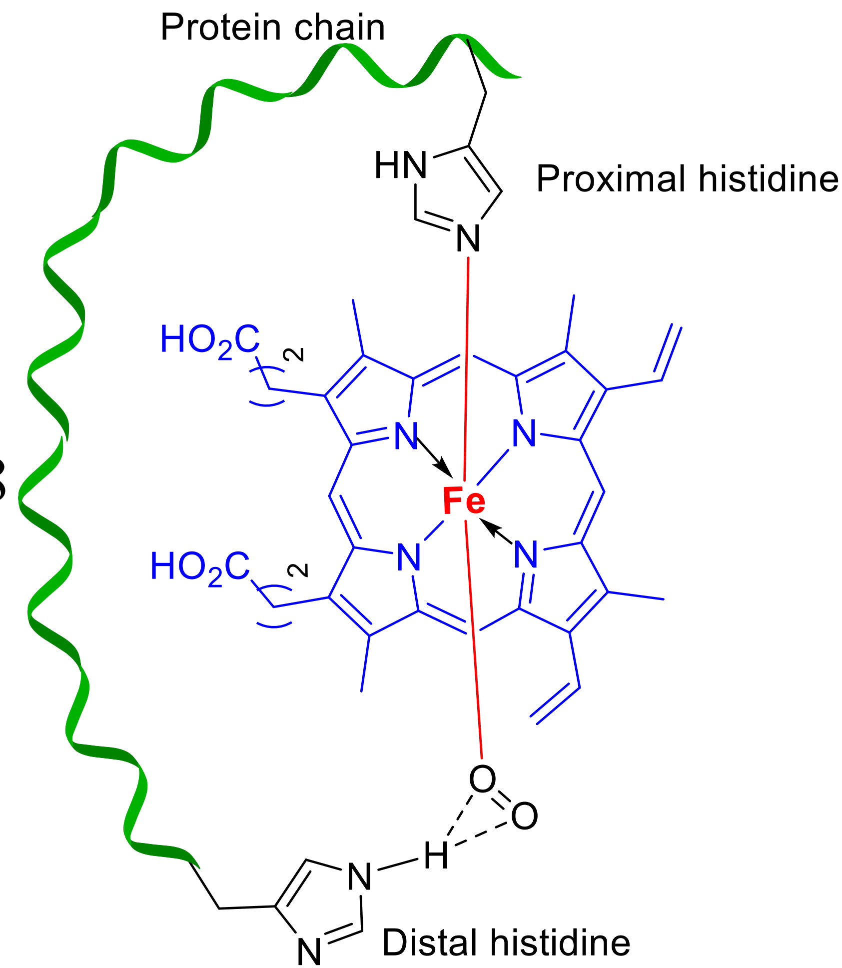
$$Hb + 4O_2 \overset{K}{\rightleftharpoons} Hb(O_2)_4$$

$$k_1 < k_2 < k_3 < k_4$$

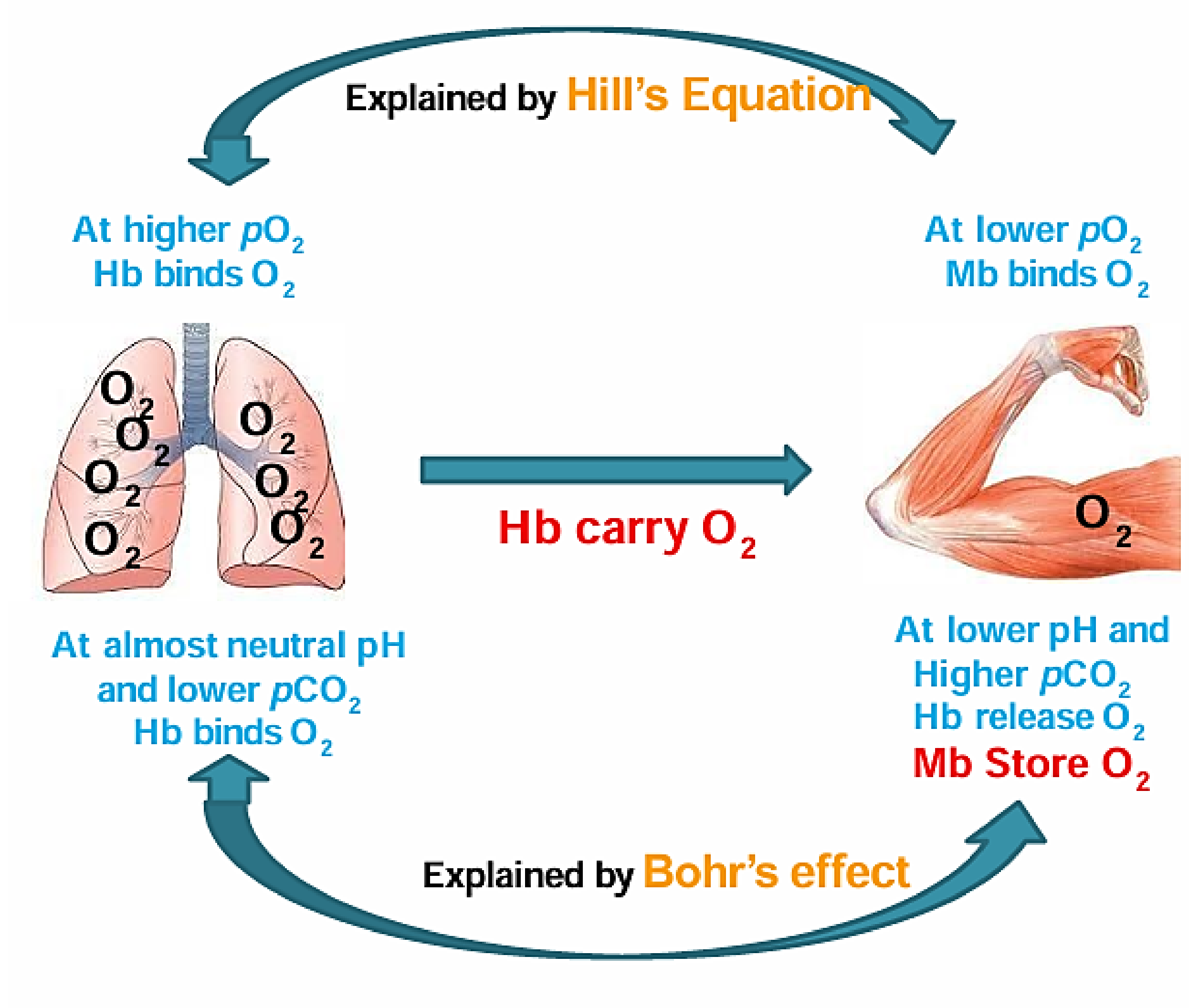
$$K = k_1 \times k_2 \times k_3 \times k_4$$

Transport of dioxygen





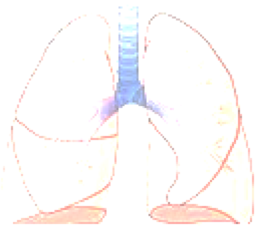
O_2 binding curve for Hb and Mb



Hill Equation

$$\theta = \frac{[\mathrm{L}]^n}{K_d + [\mathrm{L}]^n}$$

- θ is the fraction of the receptor protein concentration that is bound by the ligand,
- $[\mathrm{L}]$is the total ligand concentration,
- K_d is the apparent dissociation constant derived from the law of mass action,
- n is the Hill coefficient.



$Hb + O_2 = Hb(O_2)_4$

Hb carry O_2

Acidic

Acidic condition

$Hb(O_2)_4 = Hb + O_2$

At high O_2 pressure
High O2 affinity of Hb
Hb release CO_2

produce H_2CO_3, CO_2, lactic acid etc

Hb reverse back to lungs with CO_2

+ Mb

pH independe

Acidic condition

CO_2 transported to lungs

$Mb(O_2)$

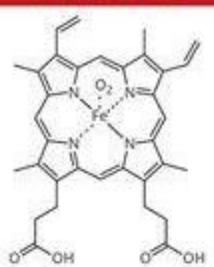
Store for respiration

THE CHEMISTRY OF THE DIFFERENT COLOURS OF BLOOD

Red

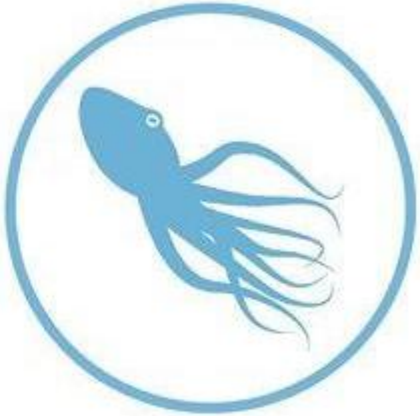
HUMANS AND THE MAJORITY OF OTHER VERTEBRATES

HAEMOGLOBIN



HAEMOGLOBIN
(oxygenated form)

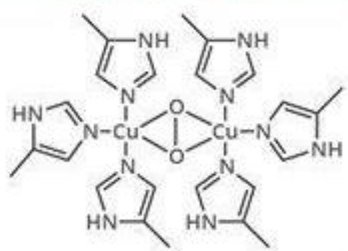
Haemoglobin is a protein found in blood, built up from subunits called 'haems'. These subunits contain iron, and their structure gives blood its red colour when oxygenated. Deoxygenated blood is a deep red colour - not blue!



Blue

SPIDERS, CRUSTACEANS, SOME MOLLUSCS, OCTOPUSES & SQUID

HAEMOCYANIN



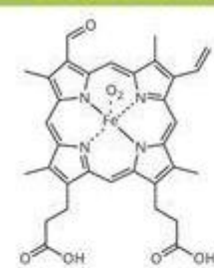
HAEMOCYANIN
(oxygenated form)

Unlike haemoglobin, which is bound to red blood cells, haemocyanin floats free in the blood. Haemocyanin contains copper instead of iron. When deoxygenated, the blood is colourless, but when oxygenated, it gives a blue colouration.

Green

SOME SEGMENTED WORMS, SOME LEECHES, & SOME MARINE WORMS

CHLOROCRUORIN



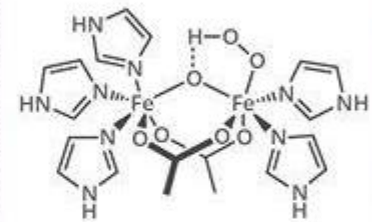
CHLOROCRUORIN
(deoxygenated form)

Chemically similar to haemoglobin; the blood of some species contains both haemoglobin & chlorocruorin. Light green when deoxygenated, it is green when oxygenated, although when more concentrated it appears light red.

Violet

MARINE WORMS INCLUDING PEANUT WORMS, PENIS WORMS & BRACHIOPODS

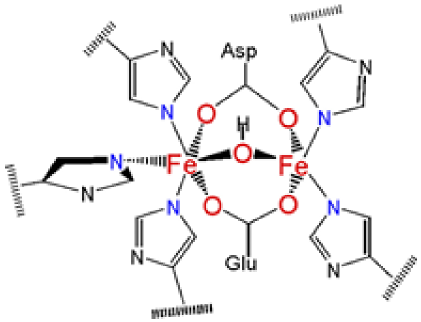
HAEMORYTHRIN



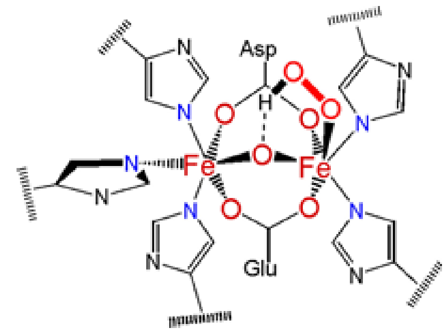
HAEMORYTHRIN
(oxygenated form)

Haemorythrin is only 1/4 as efficient at oxygen transport when compared to haemoglobin. In the deoxygenated state, haemorythtin is colourless, but it imparts a violet-pink colour when oxygenated.

Hemerythrin



Deoxyhemerythrin, h.s. Fe(II), CN = 5, 6

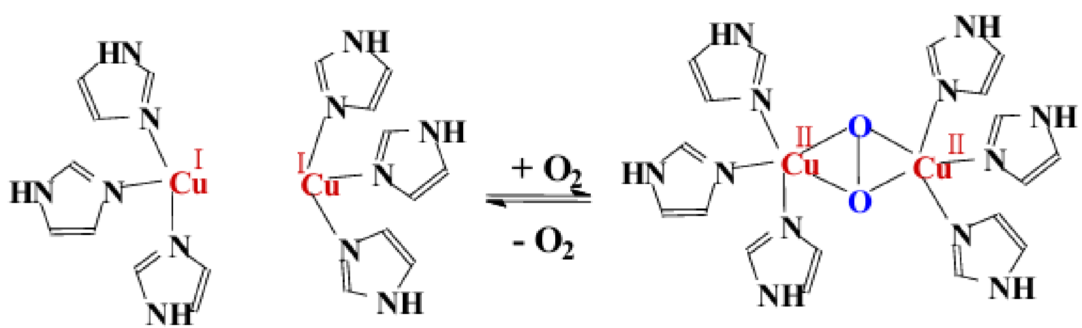


Oxyhemerythrin, Fe(III) with superexchange

- Non-porphinoid metal proteins for reversible O_2 uptake (spiders, etc.)
- Attachement of $^3O_2$ at Fe with CN = 5,
- Oxidation of both Fe (Fe(III)), reduction of $^3O_2$ to peroxide (oxo-bridge via H-transport to peroxide)

Hemocyanin

- Proteins with two Cu-centers
- Reversible oxidative addition: $O_2 \rightarrow O_2^{2-}$, $Cu^+ \rightarrow Cu^{2+}$
- Side-on bridging coordination

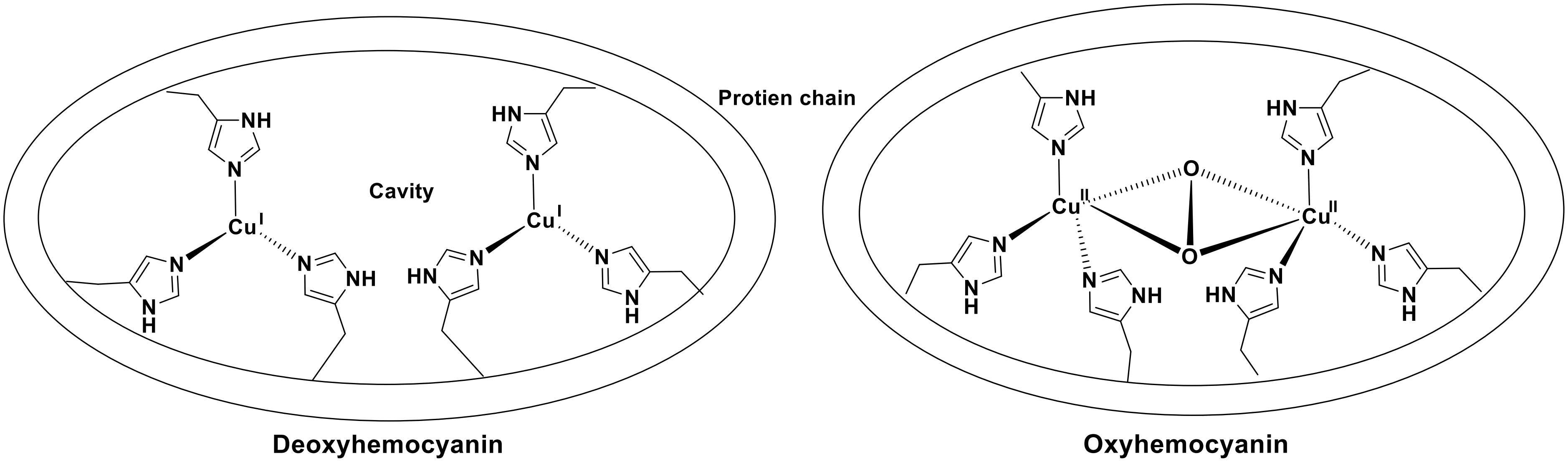


Oxygen binding in hemerythrin

- Hemerythrin holds the O_2 as a hydroperoxide
- Colorless when deoxygenated, but turn a violet-pink in the oxygenated state
- Iron atoms are bound to the protein through the carboxylate side chains of a glutamate and aspartates as well as through five histidine residues.

Hemocyanin

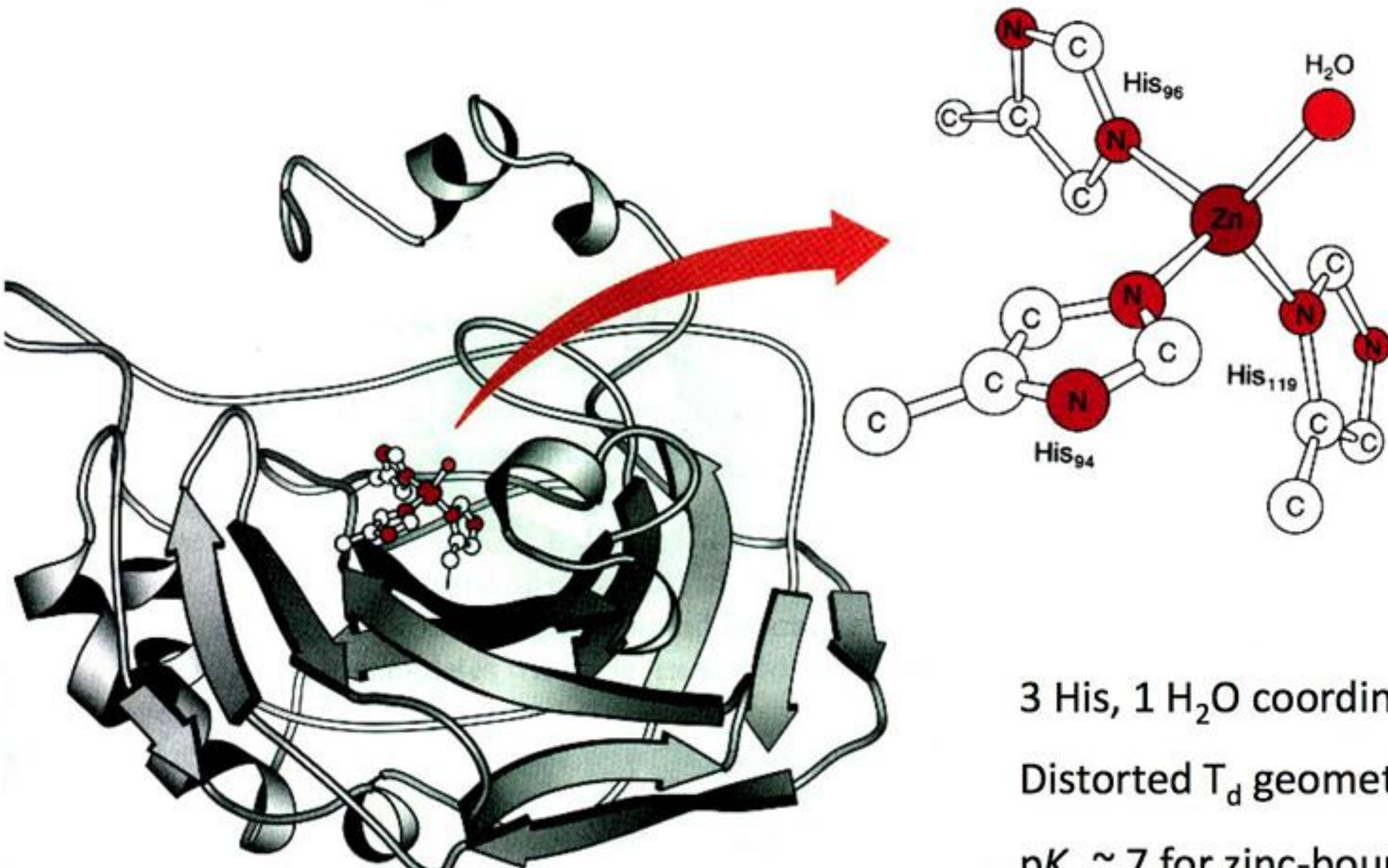
- Found in some Mollusca and Arthropoda species
- Deoxy form is colourless and oxy form is blue
- The oxygen binding center consists of two copper atoms, which exhibits +1 oxidation state in deoxy-form and +2 oxidation state in oxy-form
- Hemocyanin binds dioxygen in peroxide form and each oxygen is bonded with both copper centers.



Carbonic Anhydrase: Mononuclear Zinc Lyase

$$CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+$$

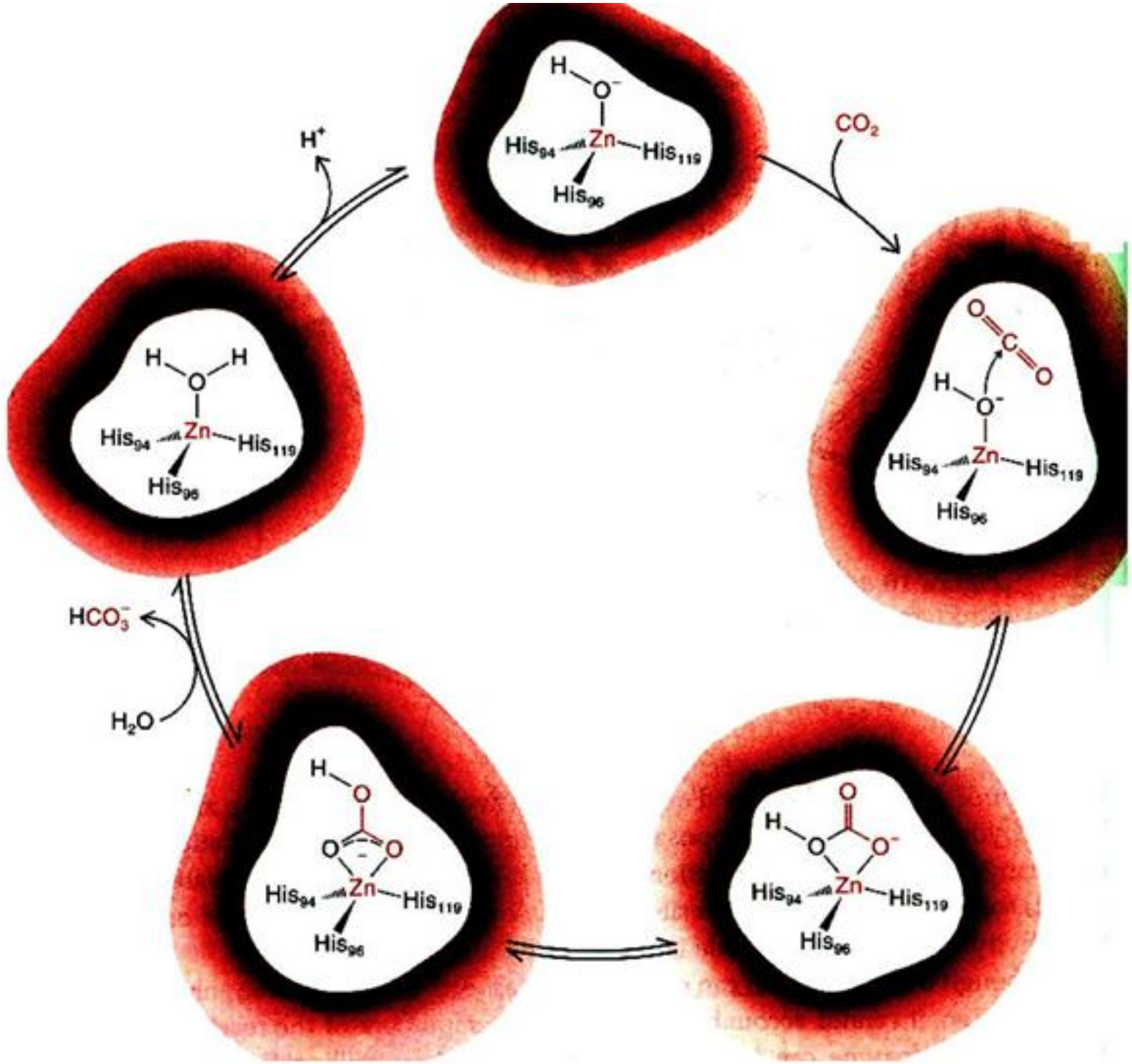
Rates: no enzyme, $10^{-2}\ s^{-1}$; enzyme, $10^{6}\ s^{-1}$



3 His, 1 H_2O coordinate Zn(II)

Distorted T_d geometry

pK_a ~ 7 for zinc-bound H_2O



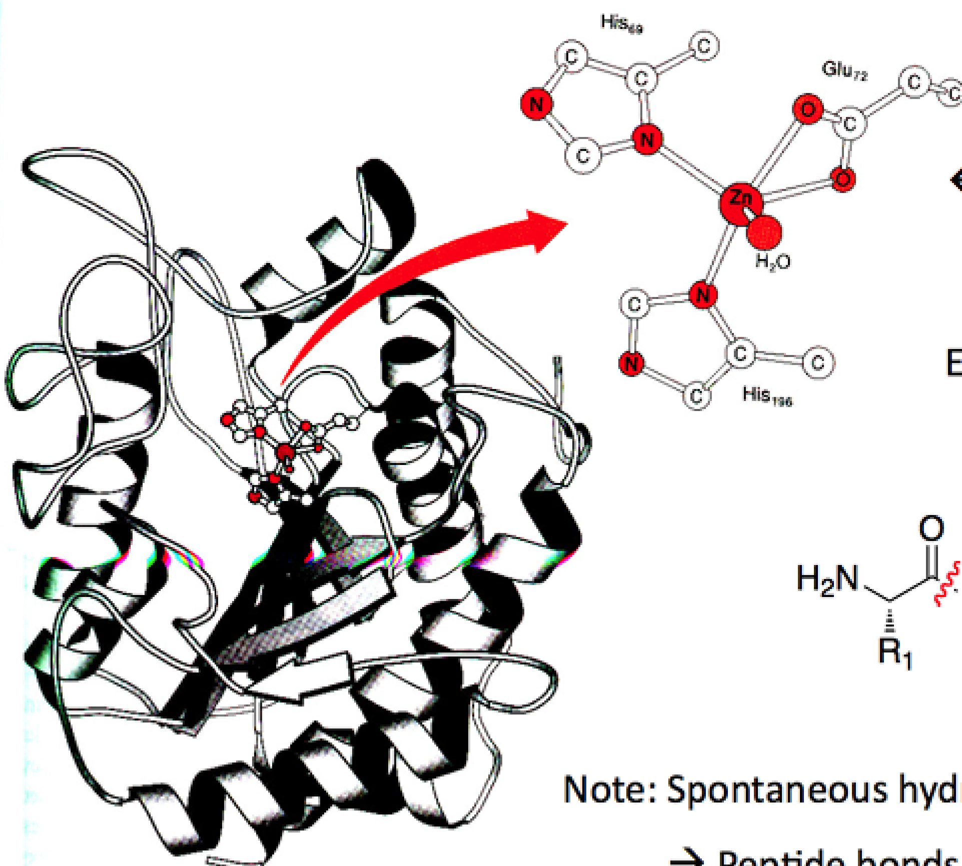
H
O⁻
His94
Zn
His119
His96
CO2
H+
H
H
O
Zn
His94
His119
His96
O
C
O
H
O⁻
His94
Zn
His119
His96
HCO3⁻
H2O
H
O
O
O
Zn
His94
His119
His96
O
H
O
O⁻
His94
Zn
His119
His96

Carboxypeptidase A

Carboxypeptidase A is a **metalloprotease** that hydrolyzes C-terminal peptide bonds

35 kDa, **pentacoordinate zinc** in the absence of substrate or inhibitor

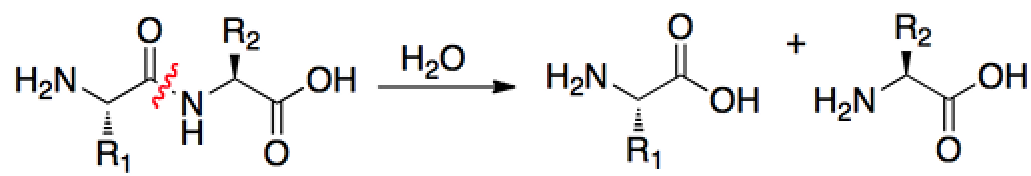
Glu_{77} undergoes **carboxylate shift** with substrate/inhibitor coordination → bi- to monodentate



← Bidentate Glu_{77}

Ex. Hydrolysis of 2-mer peptide

(aromatic & branched sidechains preferred)



Note: Spontaneous hydrolysis a pH 7 → $k = 3 \times 10^{-9}$ sec^{-1}

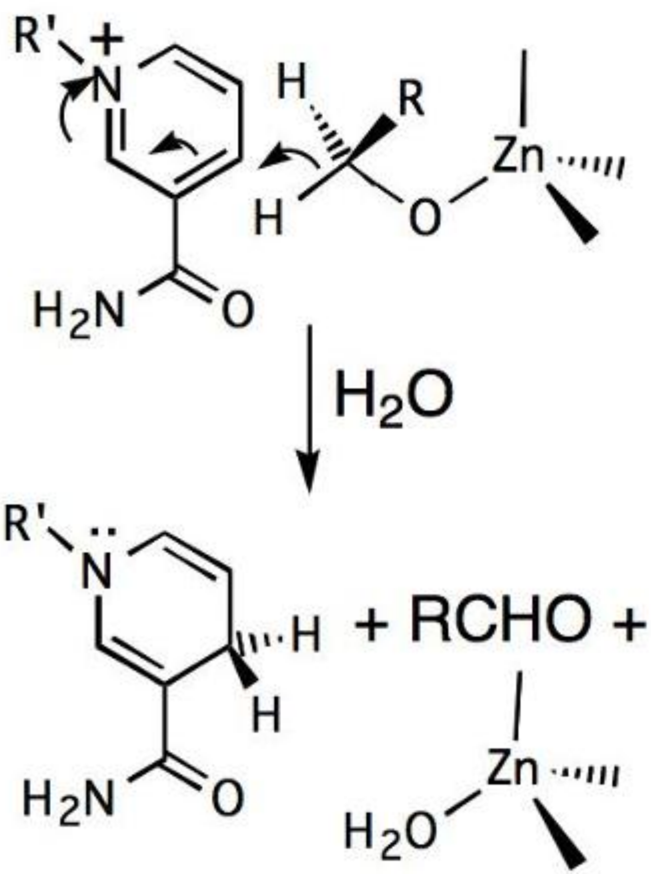
→ Peptide bonds are very stable under physiological conditions!

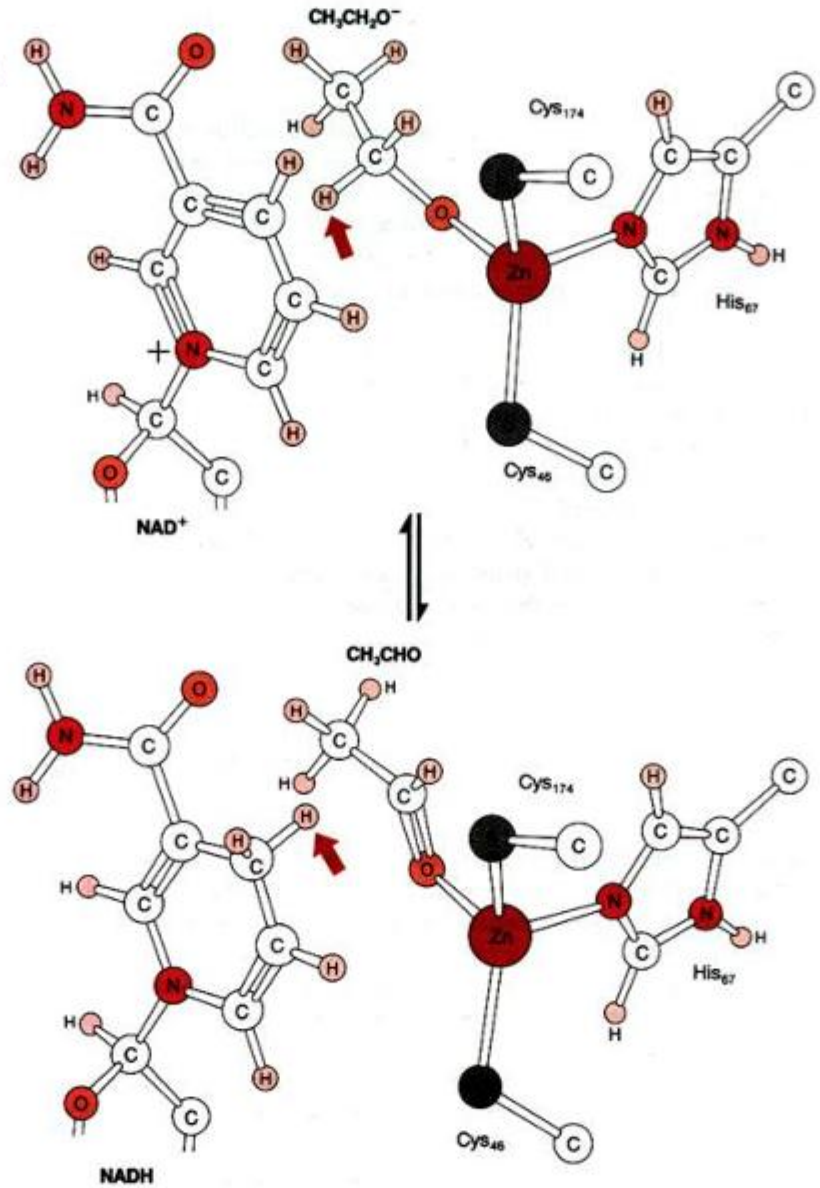
Liver Alcohol Dehydrogenase

Details of hydride transfer to NAD^+

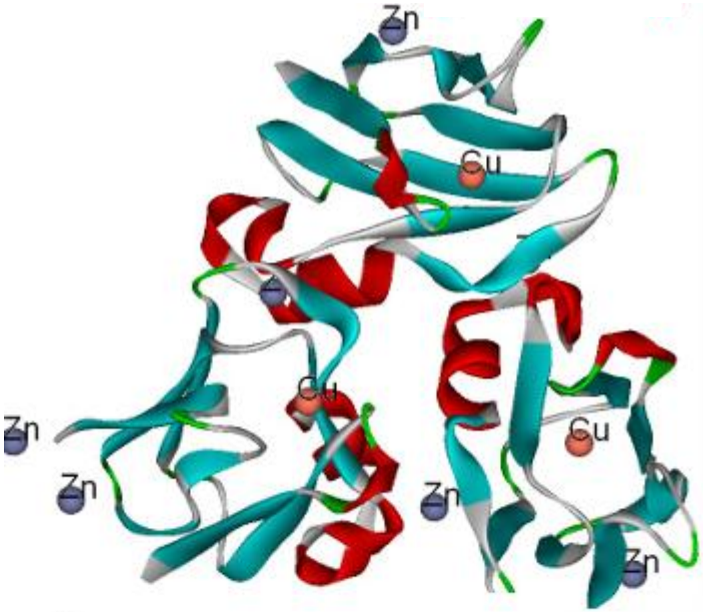
After H^- transfer, aldehyde product & NADH dissociate

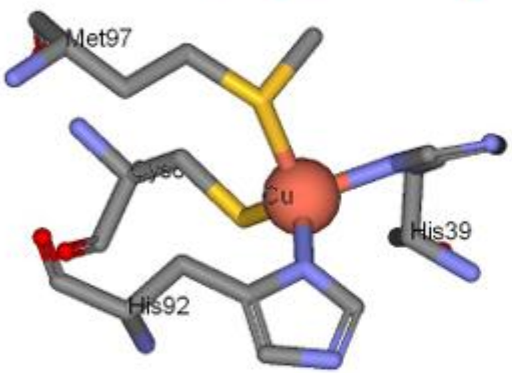
Hydride Transfer Mechanism

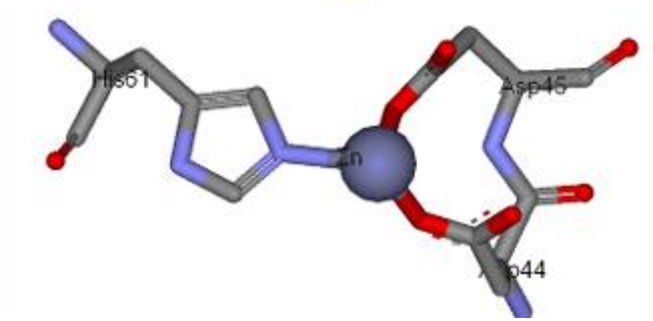




Plastocyanin







This is an electron transfer enzyme. This enzyme is able to function since Cu can undergo oxidation states of +1 and +2 easily and their inter-conversion through this protein is facile. In this enzyme, the Zn^{2+} stabilizes the protein structure that is required for the function or catalysis